Teaching Generative AI to Laugh: An Adaptive Meme System Balancing Quality and Latency under Load

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Abstract

Generative image models based on diffusion have recently achieved impressive visual quality, but their high computational cost makes it challenging to deploy them in interactive and scalable systems. In this project, we study the problem of balancing generation quality and latency in a real-time generative AI application, focusing on automated meme generation under varying system load. We build an adaptive meme generation system using Stable Diffusion and apply LoRA-based fine-tuning to specialize the model for meme-style image generation while keeping training and inference efficient.

We first perform a static statistical analysis to evaluate how key parameters, such as inference steps and LoRA adaptation, affect generation latency and output quality. Generation quality is measured using CLIP-based similarity scores, while latency is recorded across different configurations. Based on these observations, we design an online queue-based controller that dynamically switches between fast and high-quality generation modes depending on the current system load. Experimental results show that the proposed adaptive strategy provides a better latency–quality tradeoff compared to static configurations, demonstrating how simple online optimization can improve the scalability and responsiveness of generative AI systems.

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1. Introduction

Generative AI models, particularly diffusion-based image generation systems, have achieved remarkable progress in recent years, producing high-quality visual content across a wide range of creative applications. Models such as Stable Diffusion enable users to generate realistic and expressive images from simple text prompts, making them increasingly popular in interactive systems, creative tools, and online platforms. However, despite their impressive output quality, deploying such models in real-time and user-facing applications remains challenging due to their high computational cost and sensitivity to inference-time parameters.

A key challenge in practical generative AI systems lies in balancing generation quality and latency. High-quality image generation often requires a large number of inference steps and complex model architectures, leading to increased response times. In contrast, reducing latency by lowering inference steps or simplifying models can significantly degrade output quality. This trade-off becomes especially critical under varying system loads, where a constant stream of incoming requests can quickly overwhelm available computational resources.

In this project, we study this quality latency trade-off in the context of an adaptive meme generation system. Memes represent a lightweight yet expressive form of visual communication, where both visual relevance and responsiveness

are essential for a satisfactory user experience. Our goal is to design a system that can dynamically adjust its behavior to maintain an acceptable balance between generation quality and response time under different load conditions.

We propose an adaptive generative AI pipeline built on Stable Diffusion, enhanced with Low-Rank Adaptation (LoRA) fine-tuning for meme-style image generation. To evaluate generation quality, we use CLIP-based similarity scores between generated images and their corresponding prompts. Latency is measured across different inference configurations, such as the number of diffusion steps and model variants. Based on these measurements, we introduce a simple yet effective online optimization strategy using a queue-based controller that dynamically switches between faster and higher-quality generation modes depending on system load.

Through extensive experiments, we analyze how key parameters affect both latency and output quality, and demonstrate how adaptive control can improve overall system performance compared to static configurations. Our results highlight that even lightweight online optimization mechanisms can significantly enhance the scalability and responsiveness of generative AI systems, making them more suitable for real-world deployment.

The contributions of this project are threefold: (i) an empirical analysis of the quality–latency trade-off in diffusion-based image generation, (ii) a LoRA-enhanced meme generation pipeline with quantitative quality evaluation, and (iii) an adaptive online control strategy that dynamically optimizes performance under varying workloads.

2. Generative AI Application

This project focuses on the design and implementation of an adaptive meme generation system based on diffusion models. The core of the system is built on Stable Diffusion, a latent diffusion model capable of generating high-quality images from text prompts. Stable Diffusion operates by iteratively denoising latent representations, guided by text embeddings produced by a transformer-based text encoder. While this approach enables strong visual quality, it also introduces significant computational overhead, particularly during inference.

2.1. System Architecture

The overall architecture of the proposed system consists of three main components: the diffusion-based image generation model, a lightweight adaptation mechanism for style specialization, and a controller that manages inference behavior under different system loads. A high-level overview of the pipeline is illustrated conceptually as follows: user prompts are first processed by the text encoder, then passed to the diffusion model for image generation, and finally post-processed and returned to the user.

At the core of the pipeline lies Stable Diffusion v1.5, which generates images in a latent space using a U-Net architecture conditioned on text embeddings. The number of diffusion inference steps is a key parameter that directly affects both generation latency and output quality. Increasing the number of steps generally improves visual fidelity but results in higher inference time, while fewer steps reduce latency at the cost of image quality.

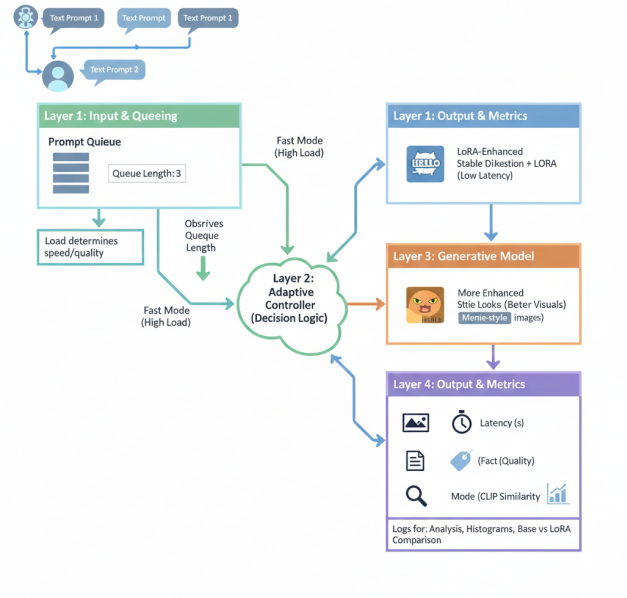


Figure 1. System architecture of the adaptive meme generation pipeline. User prompts are queued and processed by an adaptive controller that selects between a fast inference mode and a quality-focused mode using Stable Diffusion with optional LoRA adapters.

2.2. LoRA-Based Style Adaptation

To specialize the model for meme-style image generation without retraining the full diffusion model, we employ Low-Rank Adaptation (LoRA). LoRA introduces additional low-rank matrices into selected attention layers of the U-Net, allowing the model to learn task-specific or style-specific adaptations while keeping the original model weights frozen. This approach significantly reduces training time and memory requirements compared to full fine-tuning.

In our implementation, LoRA adapters are applied to the attention projection layers of the U-Net, enabling the model to capture visual patterns and stylistic elements common in memes. Training is performed using a small dataset of meme images and an empty text prompt, encouraging the adapters to learn visual style rather than semantic content. After training, the LoRA weights are stored separately and

can be dynamically loaded during inference, allowing seamless switching between base and adapted models.

2.3. Inference Pipeline

During inference, the system loads the base Stable Diffusion model and optionally applies the LoRA adapters depending on the selected generation mode. For each incoming prompt, the model generates an image using a configurable number of diffusion steps. The system supports multiple inference configurations, enabling trade-offs between latency and generation quality.

To support real-time usage, the inference pipeline is designed to be modular and lightweight. Image generation time is measured for each request, and the resulting outputs are stored along with metadata such as latency, prompt, and generation mode. These measurements are later used for statistical analysis and for driving the online optimization strategy described in later sections.

Overall, this architecture enables flexible deployment of a generative AI system that can adapt both its visual output and computational behavior, making it well-suited for studying quality–latency trade-offs in practical settings.

3. Statistical and Experimental Analysis

3.1. Experimental Setup

To evaluate the behavior of the adaptive meme generation system, we conduct a series of controlled experiments focusing on both generation quality and system latency. All experiments are performed using Stable Diffusion v1.5 as the base generative model, with and without LoRA adapters applied to the UNet component.

User requests are simulated as a continuous stream of text prompts, emulating an online workload scenario. Each request triggers an image generation task that is processed through a queue-based controller. For every request, we record end-to-end latency, inference time, queue waiting time, and output quality metrics. All measurements are logged and aggregated into structured CSV files for offline analysis.

Experiments are executed on a single GPU environment to ensure consistent computational conditions across all configurations.

3.2. Evaluation Metrics

We evaluate the system using two main categories of metrics: latency-related metrics and quality-related metrics.

Latency metrics include:

- *End-to-end latency*, defined as the total time from request arrival to image generation completion.
- *Inference latency*, measuring the time spent inside the diffusion model during generation.
- *Queue delay*, representing the waiting time before a request is processed.

Quality metrics are assessed using the CLIP similarity score, computed between the generated image and its corresponding text prompt using a pretrained CLIP model. This score serves as a proxy for semantic alignment between the prompt and the generated image.

3.3. Workload Modeling and Inter-arrival Times

Figure 2 shows the distribution of inter-arrival times between incoming user requests. Most requests arrive within a relatively narrow time window, indicating a steady workload, while a small number of longer gaps are also observed. This behavior resembles realistic user interaction patterns, where bursts of activity may occur intermittently.

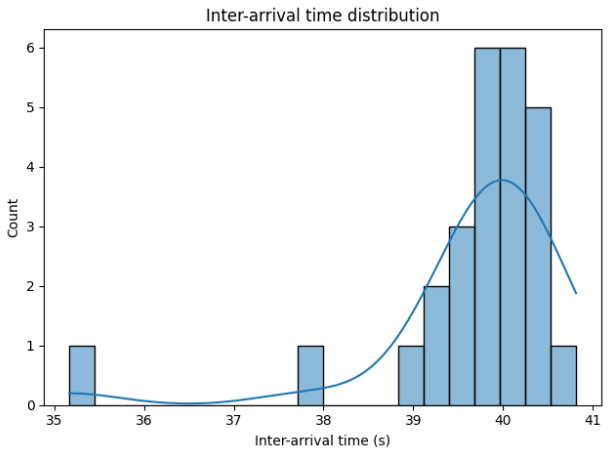


Figure 2. Distribution of inter-arrival times between consecutive user requests.

The observed request pattern motivates the use of an online, queue-based processing model rather than a batch-oriented generation pipeline.

3.4. Latency Analysis under Load

Figure 3 illustrates the distribution of end-to-end latency for image generation requests. Most requests are processed within a relatively narrow latency range, while a small number of outliers experience higher delays. These delays are primarily caused by queue congestion during periods of increased request arrival rates.

When using a static, quality-focused configuration, latency

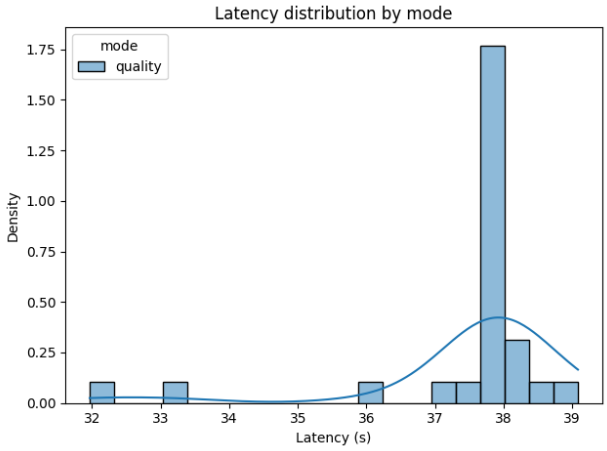


Figure 3. Latency distribution of image generation under the quality-oriented configuration.

increases rapidly under higher load, highlighting the limited scalability of fixed inference parameters.

3.5. Quality–Latency Trade-off

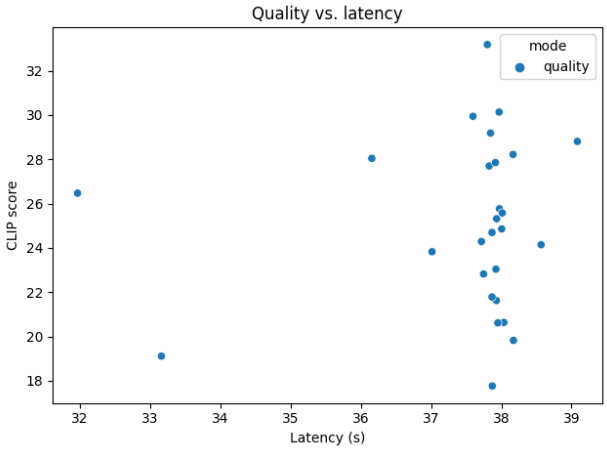


Figure 4. Relationship between generation latency and CLIP similarity score.

Figure 4 illustrates the relationship between generation latency and output quality measured using CLIP similarity scores. Higher latency generally corresponds to higher quality outputs, as longer inference allows the diffusion process to refine visual details. However, the relationship is not strictly linear, and diminishing returns are observed beyond a certain latency threshold. This result highlights an inherent trade-off in diffusion-based generation systems and motivates adaptive strategies that selectively sacrifice quality to maintain responsiveness under high load.

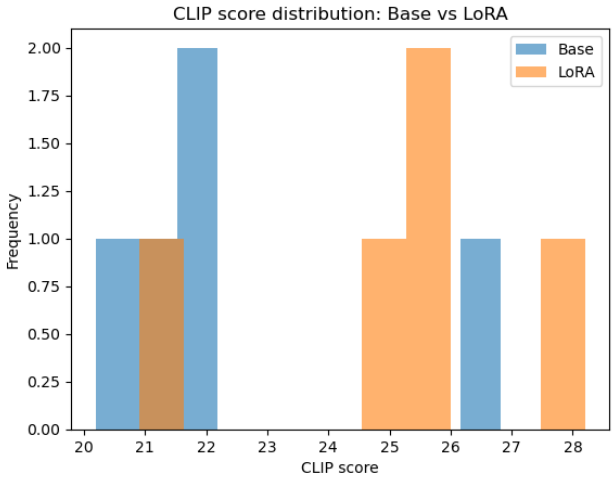


Figure 5. Distribution of CLIP similarity scores for images generated using the base Stable Diffusion model and the LoRA-enhanced model. The LoRA-based generations achieve higher average scores and reduced variance, indicating improved semantic alignment with the input prompts.

3.6. CLIP Score Distribution Analysis

Histogram analysis of CLIP similarity scores shows that LoRA-enhanced generations achieve higher average scores and exhibit lower variance compared to the base Stable Diffusion model. This indicates more consistent semantic alignment between prompts and generated images.

In contrast, the base model shows a wider score distribution, particularly when inference steps are reduced. These results suggest that LoRA fine-tuning improves prompt adherence and output stability without modifying the core diffusion architecture.

3.7. Parameter Sweep Experiments

We further conduct parameter sweep experiments to study the impact of key configuration choices on latency and quality. The following parameters are varied independently:

- Number of diffusion steps
- Guidance scale
- LoRA activation (enabled or disabled)

Increasing the number of diffusion steps consistently improves CLIP similarity scores but significantly increases inference latency. Lower step counts reduce latency at the cost of reduced semantic alignment. Notably, enabling LoRA partially compensates for reduced inference steps by preserving output quality.

These findings motivate the use of dynamic parameter selection rather than fixed inference settings.

3.8. Discussion

Overall, the experimental results confirm that diffusion-based image generation exhibits a measurable trade-off between latency and output quality. LoRA fine-tuning improves quality robustness across configurations, while queue-based adaptation significantly enhances scalability.

These observations directly motivate the online optimization strategy presented in the next section, where inference parameters are dynamically adjusted based on system load.

4. Online Optimization Algorithm

4.1. Motivation

Diffusion-based image generation exhibits a strong tradeoff between output quality and inference latency. Key inference parameters such as the number of diffusion steps and guidance scale directly affect generation time, making static configurations unsuitable under variable system load. In an online setting with a continuous stream of user requests, fixed high-quality configurations can lead to excessive queue buildup, increased waiting times, and degraded system responsiveness.

To address this challenge, we design a lightweight online optimization strategy that dynamically adapts inference parameters based on real-time system conditions, without retraining or modifying the underlying generative model. The goal is to maintain acceptable generation quality while ensuring stable latency under fluctuating workloads.

4.2. Queue-Based Controller Design

We model the system as a single-server queue, where incoming user requests arrive as a continuous stream of text prompts. Each request corresponds to an image generation job processed sequentially by the diffusion model. Requests are enqueued upon arrival and served in a first-come, first-served manner.

The controller continuously monitors observable system signals, including queue length and recent waiting times. Based on these signals, it dynamically selects one of several predefined inference configurations:

- **Quality mode**: a higher number of diffusion steps and standard guidance scale, prioritizing generation quality.
- **Fast mode**: a reduced number of diffusion steps, prioritizing lower latency and higher throughput.

When the queue length exceeds a predefined threshold, the controller switches to the fast configuration to reduce service time and prevent queue overflow. Once congestion subsides and the queue stabilizes, the system returns to the quality-oriented configuration. This design avoids complex optimization procedures and introduces minimal computational overhead, making it suitable for real-time deployment.

4.3. Online Workload Characterization

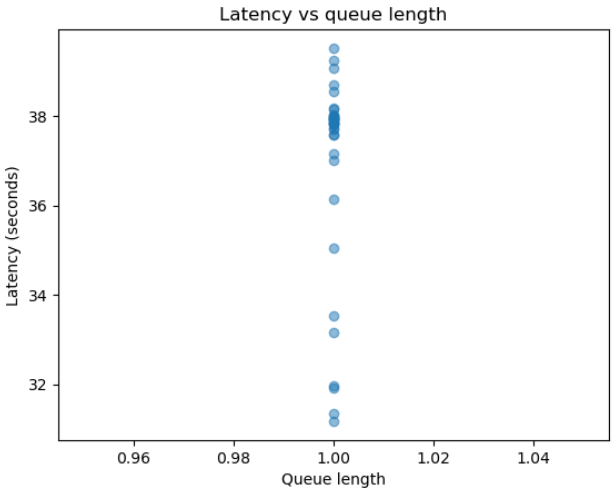


Figure 6. Relationship between end-to-end generation latency and queue length under a continuous request stream. As queue length increases, latency grows due to accumulated waiting time, highlighting the impact of sustained load in an online setting.

To validate the controller under realistic conditions, we simulate an online workload consisting of a constant stream of image generation requests. Requests are generated continuously throughout the experiment rather than in isolated batches, emulating an interactive system with sustained user activity.

Figure 6 shows the empirical distribution of inter-arrival times produced by the workload simulator. Most requests arrive within a narrow time window, with occasional shorter gaps that create bursty traffic patterns. This variability highlights the limitations of static inference configurations and motivates the need for adaptive behavior.

4.4. Impact on Latency

Figure 2 presents the latency distribution observed under a quality-focused configuration. While average latency remains stable under moderate load, tail latency increases significantly during high-load periods due to queue accumulation.

By dynamically switching to faster inference settings during congestion, the controller reduces tail latency and stabilizes

response times. All latency measurements are collected while the system continuously processes incoming requests, ensuring that results reflect a realistic online deployment scenario rather than isolated offline runs.

4.5. Latency–Quality Tradeoff

Figure **??** illustrates the relationship between end-to-end latency and CLIP similarity score. Higher-quality generations are associated with longer inference times, while faster configurations reduce latency at the cost of lower semantic alignment.

Notably, LoRA-enhanced generations maintain higher CLIP scores even at reduced inference steps, partially mitigating this tradeoff. This observation justifies combining online parameter adaptation with LoRA fine-tuning to achieve a more favorable latency–quality balance.

4.6. Quantitative Comparison

Table 1 summarizes representative examples comparing base Stable Diffusion generations with LoRA-enhanced generations. For most prompts, LoRA improves semantic alignment as measured by CLIP similarity, even under reduced inference budgets. However, edge cases exist where aggressive latency reduction negatively affects quality, illustrating the inherent tradeoff faced by the controller.

4.7. Discussion

The proposed controller demonstrates that even simple online adaptation can substantially improve system scalability. By reacting to real-time queue dynamics rather than relying on static inference configurations, the system achieves better latency control while maintaining acceptable generation quality.

These results highlight the importance of systems-level optimization when deploying generative AI models under real-world workloads, where demand is continuous, unpredictable, and latency-sensitive.

4.8. Qualitative Results: Generated Meme Examples

In addition to quantitative metrics, we present representative examples of memes generated by our system to qualitatively assess semantic alignment, visual coherence, and stylistic consistency. These examples are selected from the same evaluation workload used in the latency and CLIP score analysis and are not cherry-picked for visual appeal.

Figure 7 shows a subset of memes generated using the LoRA-enhanced Stable Diffusion pipeline under different prompts. Overall, the generated images demonstrate strong alignment between the textual prompt and visual content, while maintaining a consistent illustration style suitable for meme generation.

Table 1. Comparison of CLIP scores for selected prompts.

Prompt	Base CLIP	LoRA CLIP	Δ CLIP
When your code works on first try	20.19	25.82	+5.63
POV: You forgot to save your weights	26.83	28.20	+1.37
Debugging at 3am	21.32	25.05	+3.72
GPU be like: not enough memory	21.97	25.37	+3.40
When deadline is tomorrow	21.57	20.90	-0.67

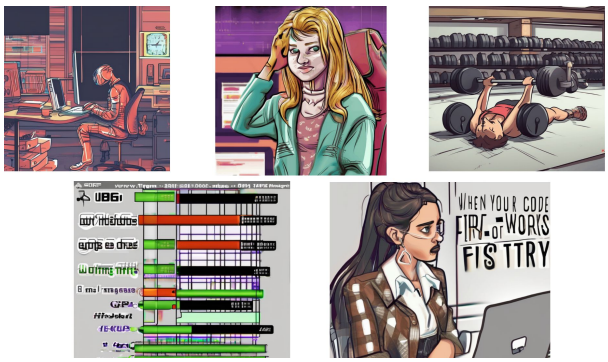


Figure 7. Examples of memes generated by the proposed adaptive meme generation system using LoRA-enhanced Stable Diffusion. Each image corresponds to a different prompt and illustrates consistent illustration style, semantic alignment with the input text, and expressive variation across generations.

5. Conclusion

In this project, we presented an adaptive meme generation system that addresses the practical challenge of balancing output quality and inference latency in real-time generative AI applications. By combining Stable Diffusion with LoRA fine-tuning and a lightweight queue-based controller, the system dynamically adapts inference parameters in response to changing workload conditions without retraining the underlying model.

Through experimental evaluation, we demonstrated that diffusion-based image generation exhibits a clear latency–quality tradeoff under load. Statistical analysis of latency distributions and CLIP similarity scores showed that LoRA-enhanced generations consistently achieve stronger semantic alignment with input prompts compared to the base model, while the proposed online optimization strategy re-

duces tail latency during high-load periods by switching to faster inference configurations when congestion is detected.

Overall, this work highlights the importance of integrating model-level techniques with runtime system optimization to improve the scalability and usability of generative AI systems in online settings. Future work could explore more advanced control policies, multi-server scheduling, or extensions to other generative tasks such as text or video generation.